25901275 **Reg.No : .....**

Name : .....

MAHATMA GANDHI UNIVERSITY, KOTTAYAM

MGU-UGP (HONOURS) REGULAR EXAMINATION March 2026

Fourth SEMESTER

Discipline Specific Elective (DSE) - MG4DSEFTQ201 - FOOD ENGINEERING

(2024 ADMISSION ONWARDS)

Duration: 2 hours **Maximum Marks: 70**

Remember(K), Understand(U), Apply(A), Analyse(An), Evaluate(E), Create(C), Skill(S), Interest(I) and Appreciation(Ap)

Students should attempt at least one question from each course outcome to enhance their overall outcome attainability.

Part A

Short Answer type

Answer any **10** questions

Each question carries **2** marks

1. Define thermal diffusivity [K] / [CO2]
2. Define fundamental units. [K] / [CO1]
3. How to convert kelvin to celcius scale ? [A] / [CO1]
4. Viscosity an important property in food processing operations. Justify [E] / [CO4]
5. Write mass balance equation for steady state system. [U] / [CO1]
6. Differentiate between coarse crushing and fine grinding. [U] / [CO5]
7. What happens to water during freezing? [U] / [CO3]
8. Define ohmic heating. [U] / [CO2]
9. Comment on lyophilization. [K] / [CO5]
10. Define rheology. [K] / [CO4]
11. State the methods to reduce water activity in foods. [K] / [CO3]

12. What is the condition for an adiabatic process? [U] / [CO3]

[2x10 = 20]

Part B

Short Essay Questions

Answer any **6** questions

Each question carries **5** marks

13. Describe the construction and working of a scraped surface heat exchanger. [U] / [CO2]

14. Describe surface tension and explain its importance in food processing operations. [U] / [CO4]

15. Write a short note on equipments used for size reduction of liquid foods. [A] / [CO5]

16. Explain rotary pumps. [U] / [CO4]

17. Differentiate First and Second Laws of Thermodynamics. [An] / [CO3]

18. Write a short note on emissivity and its role in radiation heat transfer. [An] / [CO2]

19. Explain U-tube manometer. [U] / [CO4]

20. Compare pasteurization and sterilization based on thermodynamic principles. [An] / [CO3]

[5x6 = 30]

Part C

Long Essay Questions

Answer any **2** questions

Each question carries **10** marks

21. Explain basic drying process with drying rate curve and stages. [An] / [CO5]

22. Discuss the concepts of internal energy and enthalpy and explain their significance in food processing operations. [An] / [CO3]

23. Compare natural convection and forced convection with respect to mechanism, heat transfer coefficient, and applications in food processing. [U] / [CO2]

24. Derive the continuity equation. [C] / [CO4]

[10x2 = 20]